

Module 01: Assignment 01 - Exercise 01: Running Data Regression & Clustering Exercise

Nischal Pant

1. Explore the data. What is the relationship between Heartrate, Cadence, and Elevation? Are there other meaningful relationships?

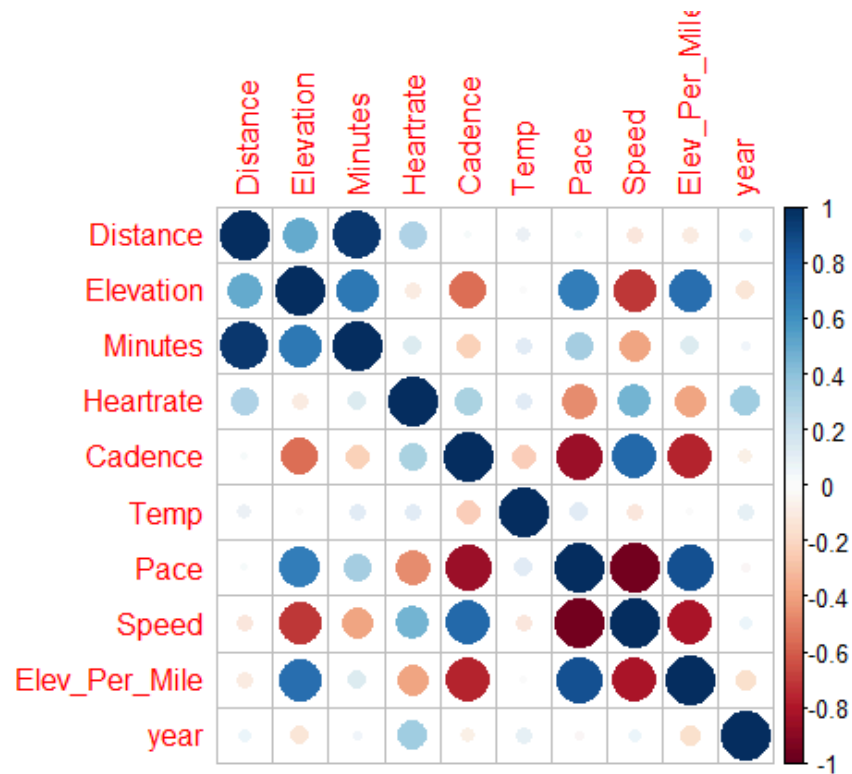
We can use the following code to analyze the correlation between the different variables in our run dataset. Looking at the result of the cut-off of 0.7 we can see which variables are correlated. To see if there are also negative correlations i used the code to find absolute values. We can see that they are correlated by the plot produced. I would not consider others too be significant as their correlation is still lower than 0.7.

```
```{r}

Calculate correlations between numeric variables
cor_df <- df %>%
 select_if(is.numeric)

c <- round(cor(cor_df), digits = 2)
c
Visualize correlation matrix
corrplot::corrplot(c)

Find highly correlated measures
highCorr <- findCorrelation(c, cutoff = 0.7, names = TRUE)
highCorr
```
```



```
[1] "Speed" "Pace" "Elevation" "Elev Per Mile" "Minutes"
```

```
# Find highly correlated measures (including both positive and negative correlations)
highCorr ← findCorrelation(abs(c), cutoff = 0.7, names = TRUE)
highCorr

# Find moderately correlated measures
midCorr ← findCorrelation(abs(c), cutoff = 0.2, names = TRUE)
midCorr
```

```
"Speed" "Pace" "Elevation" "Elev Per Mile" "Cadence" "Minutes" "Heartrate"
```

This is the correlation between heart rate, Cadence and Elev_Per_Mile

```

{r}

d <- run.data %>%
  filter(!Area %in% "Treadmill") %>%
  select(Date:Terrain, Pace:Elev_Per_Mile, PE, year)

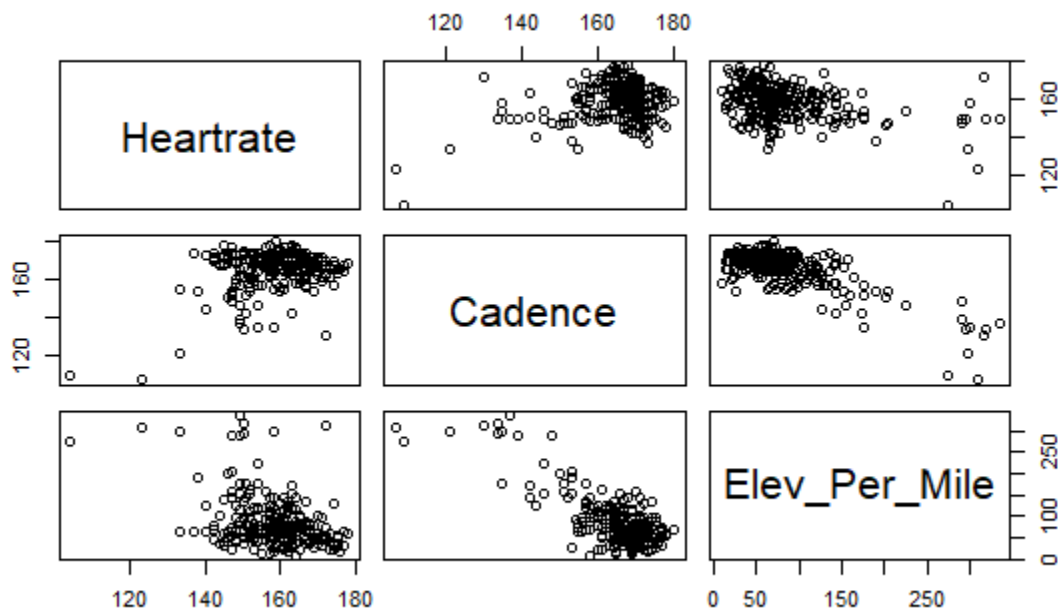
d <- d[complete.cases(d[, c("Heartrate", "Cadence", "Elev_Per_Mile")]), ]

pairs(d[, c("Heartrate", "Cadence", "Elev_Per_Mile")])

plot_matrix <- ggplot(d, aes(x = Heartrate, y = Cadence)) +
  geom_point() +
  facet_grid(~Elev_Per_Mile) +
  labs(title = "Scatter Plot Matrix", x = "Heartrate", y = "Cadence")

print(plot_matrix)

```



2. Improve the accuracy of the current model on pace.

This is the current result of the model on pace

```
Call:
lm(formula = formula, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-3.1412 -0.4476 -0.0629  0.5318  3.5377

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    35.119408   1.841194   19.074 < 2e-16 ***
Distance         0.124409   0.028813    4.318 2.26e-05 ***
Heartrate       -0.049080   0.006483   -7.570 6.81e-13 ***
Cadence         -0.119280   0.007722  -15.448 < 2e-16 ***
Temp             0.003453   0.004066    0.849  0.396
log(Elev_Per_Mile):Terrain0  0.697085   0.129390    5.387 1.62e-07 ***
log(Elev_Per_Mile):Terrain1  0.858284   0.118369    7.251 4.94e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8676 on 255 degrees of freedom
Multiple R-squared:  0.838,    Adjusted R-squared:  0.8342
F-statistic: 219.8 on 6 and 255 DF,  p-value: < 2.2e-16
```

| | GVIF | Df | GVIF^(1/(2*Df)) |
|----------------------------|----------|----|-----------------|
| Distance | 1.123077 | 1 | 1.059753 |
| Heartrate | 1.314210 | 1 | 1.146390 |
| Cadence | 2.106763 | 1 | 1.451469 |
| Temp | 1.180915 | 1 | 1.086699 |
| log(Elev_Per_Mile):Terrain | 2.045909 | 2 | 1.195974 |

Things we can do to improve the pace.

Add interaction terms

```
```{r}
Add interaction terms
d$InteractionTerm1 <- d$Distance * d$Heartrate
d$InteractionTerm2 <- d$Cadence * d$Temp

improved_lm <- fit_lm(d, Pace ~ Distance + log(Elev_Per_Mile) + Heartrate +
Cadence + Temp + Terrain +
 log(Elev_Per_Mile):Terrain + InteractionTerm1 +
InteractionTerm2)|
improved_lm
```
```

```
$summary

Call:
lm(formula = formula, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-2.85611 -0.43664 -0.04741  0.49956  2.54092

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.173e+01  4.661e+00   2.517 0.012443 *
Distance        1.808e-01  4.458e-01   0.406 0.685409
log(Elev_Per_Mile)  5.044e-01  1.211e-01   4.166 4.27e-05 ***
Heartrate       -4.401e-02  1.252e-02  -3.515 0.000522 ***
Cadence         1.794e-02  2.467e-02   0.727 0.467743
```

Identify and remove outliers

```
```{r}

outliers <- boxplot.stats(d$Pace)$out
d <- d[!d$Pace %in% outliers,]

improved_lm <- fit_lm(d, Pace ~ Distance + log(Elev_Per_Mile) + Heartrate +
Cadence + Temp + Terrain +
 log(Elev_Per_Mile):Terrain + InteractionTerm1 +
InteractionTerm2)|
improved_lm
```
```

\$summary

Call:

lm(formula = formula, data = data)

Residuals:

| Min | 1Q | Median | 3Q | Max |
|----------|----------|----------|---------|---------|
| -2.14128 | -0.34132 | -0.05874 | 0.39902 | 2.57014 |

Coefficients:

| | Estimate | Std. Error | t value | Pr(> t) | |
|-----------------------------|------------|------------|---------|----------|-----|
| (Intercept) | 18.2448052 | 5.6450072 | 3.232 | 0.00141 | ** |
| Distance | -0.1857327 | 0.4235088 | -0.439 | 0.66139 | |
| log(Elev_Per_Mile) | 0.5187643 | 0.1056825 | 4.909 | 1.72e-06 | *** |
| Heartrate | -0.0537295 | 0.0126518 | -4.247 | 3.13e-05 | *** |
| Cadence | -0.0126104 | 0.0300576 | -0.420 | 0.67521 | |
| Temp | 0.0956228 | 0.0856402 | 1.117 | 0.26533 | |
| Terrain1 | -3.7295441 | 1.8739650 | -1.990 | 0.04773 | * |
| InteractionTerm1 | 0.0022514 | 0.0025898 | 0.869 | 0.38555 | |
| InteractionTerm2 | -0.0005329 | 0.0005097 | -1.045 | 0.29692 | |
| log(Elev_Per_Mile):Terrain1 | 1.0088803 | 0.4051255 | 2.490 | 0.01346 | * |

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6718 on 234 degrees of freedom

Multiple R-squared: 0.6738, Adjusted R-squared: 0.6612

F-statistic: 53.7 on 9 and 234 DF, p-value: < 2.2e-16

\$vif

| | |
|----------------------------|--------------------|
| Distance | log(Elev_Per_Mile) |
| 382.552162 | 1.580125 |
| Heartrate | Cadence |
| 6.048087 | 16.919215 |
| Temp | Terrain |
| 826.751914 | 285.609624 |
| InteractionTerm1 | InteractionTerm2 |
| 410.265035 | 782.755753 |
| log(Elev_Per_Mile):Terrain | |
| 296.206568 | |

Apply log transformation

```
``{r}  
  
d$logHeartRate <- log(d$HeartRate + 1) # Adding 1 to handle zero values  
improved_lm <- fit_lm(d, Pace ~ Distance + log(Elev_Per_Mile) + logHeartRate  
+ Cadence + Temp + Terrain +  
                        log(Elev_Per_Mile):Terrain + InteractionTerm1 +  
InteractionTerm2)|  
improved_lm  
``
```

\$summary

Call:

lm(formula = formula, data = data)

Residuals:

| Min | 1Q | Median | 3Q | Max |
|----------|----------|----------|---------|---------|
| -2.12555 | -0.34397 | -0.05904 | 0.39250 | 2.58476 |

Coefficients:

| | Estimate | Std. Error | t value | Pr(> t) | |
|-----------------------------|------------|------------|---------|----------|-----|
| (Intercept) | 52.0683862 | 11.4294077 | 4.556 | 8.41e-06 | *** |
| Distance | -0.1456658 | 0.4106273 | -0.355 | 0.7231 | |
| log(Elev_Per_Mile) | 0.5248277 | 0.1054784 | 4.976 | 1.26e-06 | *** |
| logHeartRate | -8.3631138 | 1.9424945 | -4.305 | 2.45e-05 | *** |
| Cadence | -0.0124147 | 0.0300233 | -0.414 | 0.6796 | |
| Temp | 0.0945508 | 0.0855697 | 1.105 | 0.2703 | |
| Terrain1 | -3.7000042 | 1.8724666 | -1.976 | 0.0493 | * |
| InteractionTerm1 | 0.0020066 | 0.0025102 | 0.799 | 0.4249 | |
| InteractionTerm2 | -0.0005261 | 0.0005093 | -1.033 | 0.3027 | |
| log(Elev_Per_Mile):Terrain1 | 1.0030381 | 0.4047882 | 2.478 | 0.0139 | * |

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6711 on 234 degrees of freedom

Multiple R-squared: 0.6744, Adjusted R-squared: 0.6619

F-statistic: 53.86 on 9 and 234 DF, p-value: < 2.2e-16

\$vif

| | Distance | log(Elev_Per_Mile) |
|--|----------------------------|--------------------|
| | 360.349065 | 1.577155 |
| | logHeartRate | Cadence |
| | 5.668802 | 16.914056 |
| | Temp | Terrain |
| | 827.030535 | 285.719566 |
| | InteractionTerm1 | InteractionTerm2 |
| | 386.202490 | 783.056141 |
| | log(Elev_Per_Mile):Terrain | |
| | 296.300920 | |

Use cross-validation for model evaluation

```
```{r}
control_cv <- trainControl(method = "cv", number = 10)
improved_lm_cv <- train(Pace ~ Distance + log(Elev_Per_Mile) + logHeartrate +
 Cadence + Temp + Terrain +
 log(Elev_Per_Mile):Terrain + InteractionTerm1 +
 InteractionTerm2,
 data = d, method = "lm", trControl = control_cv)
summary(improved_lm_cv)
```
```

```
Call:
lm(formula = .outcome ~ ., data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-2.12555 -0.34397 -0.05904  0.39250  2.58476

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   52.0683862   11.4294077    4.556 8.41e-06
Distance     -0.1456658    0.4106273   -0.355  0.7231
`log(Elev_Per_Mile)`
  0.5248277    0.1054784    4.976 1.26e-06
logHeartrate  -8.3631138    1.9424945   -4.305 2.45e-05
Cadence       -0.0124147    0.0300233   -0.414  0.6796
Temp           0.0945508    0.0855697    1.105  0.2703
Terrain1      -3.7000042    1.8724666   -1.976  0.0493
InteractionTerm1
  0.0020066    0.0025102    0.799  0.4249
InteractionTerm2
 -0.0005261    0.0005093   -1.033  0.3027
`log(Elev_Per_Mile):Terrain1`
  1.0030381    0.4047882    2.478  0.0139

(Intercept)          ***
Distance              ***
`log(Elev_Per_Mile)`  ***
logHeartrate          ***
Cadence
Temp
Terrain1              *
InteractionTerm1
InteractionTerm2
`log(Elev_Per_Mile):Terrain1` *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6711 on 234 degrees of freedom
Multiple R-squared:  0.6744,    Adjusted R-squared:  0.6619
F-statistic: 53.86 on 9 and 234 DF,  p-value: < 2.2e-16
```

Adding Interaction Terms:

Interaction terms allow the model to capture the combined effect of two or more variables, providing a more realistic representation of the relationship between predictors and the target variable.

For example, if the effect of one variable on the target variable depends on the value of another variable, adding an interaction term can help capture this dependency.

Removing Outliers:

Outliers can disproportionately influence the regression model, leading to biased parameter estimates.

By removing outliers, the model tends to become more robust and less sensitive to extreme data points, resulting in a more accurate representation.

Log Transformations:

Log transformations are useful when the relationship between predictors and the target variable is not linear. Applying log transformations to skewed or exponentially distributed variables can make their relationship with the target more linear, improving the model's ability to capture patterns in the data. In this case, we logged the heart rate variable.

Cross-Validation:

By splitting the dataset into multiple training and testing subsets, cross-validation helps ensure that the model generalizes well to new data, reducing the risk of overfitting.

Overall Improvements:

The combined effect of these techniques is likely to result in a more accurate and precise regression model. Interaction terms, removal of outliers, and log transformations can help capture complex relationships, while cross-validation ensures that the model performs well on new, unseen data. By using cross-validation and removing outliers, the model is less likely to overfit the training data. Overfitting occurs when a model is too complex and fits noise in the data rather than the underlying patterns. Adding interaction terms and transforming variables can lead to a more interpretable model that better reflects the real-world relationships among variables.

5. With unfiltered data (i.e., don't remove treadmill data), create a clustering model to backfill perceived exertion. Write a short summary explaining how and why it works to a non-technical audience. Be sure to test the performance of your model if you can.

This code is performing hierarchical clustering on a dataset containing various running-related variables. We can break it down by the following to create a summary for a non-technical audience.

- **Data Preprocessing:** We will first select all the relevant columns that we require for the clusters. We can now remove rows with missing values in the 'cadence,' 'PE,' and 'elevation' columns. It

converts the 'Terrain' column to numeric. Missing values in the 'Temp' column are imputed with 65, and missing values in the 'Terrain' column are imputed with 0.

- **Scaling and Hierarchical Clustering:** The code then converts selected features into a matrix and scales the features. It calculates the Euclidean distances between observations. Hierarchical clustering is performed using the complete linkage method.
- **Visualizing Clusters:** A dendrogram is plotted, and a horizontal line is drawn to cut the dendrogram into six clusters.
- **Cluster Assignment:** The clusters are assigned to each observation, and this information is added to the dataframe.
- **Summary of Clusters:** The code generates a summary of each cluster, calculating the mean values of various features within each cluster.
- **Analysis of Perceived Exertion (PE):** The code analyzes the distribution of PE within each cluster, providing a count of occurrences for each PE value. Now, let's explain how and why this works in a non-technical manner:

Summary for a Non-Technical Audience:

We used a data-driven approach to group similar running sessions into clusters, considering factors like distance, elevation, heart rate, cadence, temperature, and terrain. This allows us to identify patterns and similarities in our running data. Each cluster represents a group of runs with similar characteristics. We then assigned each run to a cluster based on these characteristics.

This clustering approach helps us understand the different types of runs in our dataset. We found six distinct clusters, each with its own average values for distance, elevation, time, heart rate, cadence, temperature, terrain, and pace. This allows us to categorize and analyze our runs more effectively.

To enhance our dataset, we applied this clustering information to fill in missing values, especially for perceived exertion (PE). For example, if the PE value was missing for a run, we estimated it by looking at the average PE value within the cluster to which that run belongs.

In essence, this method leverages the similarities among runs to make reasonable estimates for missing information. It helps us make the most of our running data and gain insights into different running experiences.